Application: Interpolation

Suppose that for 10 distinct experimental input values $t_0, t_1, \cdots, t_9$, we obtained the 10 output values y_0, y_1, $\cdots$ y_9. We would like to find a precise functional relationship between y_i and t_i, $i = 0, 1, \cdots, 9$. That is we would like to find a continuous function $y = f(t)$ such that $f(t_i) = y_i$. One of the reason is the following. For example, if t_i denotes the ith year and y_i denotes the sales of a company at the end of the ith year, we have sales record only at the end of $0^{th}, 1^{st}, 2^{nd}, \cdots, 9^{th}$ years. Suppose we want the sales record at the two and a half years or at the end of 9th month of the 6th year. The function $y = f(t)$ will help us find it. The process to find $y = f(t)$ is called **interpolation**.

Most of the times, the relationship between t_i and y_i is not linear, that is as t increases, $y = f(t)$ is either completely increasing or decreasing. This is true for sales, it is never uniform. Hence, most of the times $y = f(t)$ as a linear fitting of the data is not correct. Geometrically, linear fitting means a straight line $y = mt + c$ which passes through the the points (t_i, y_i). Instead, we look for a non-linear fitting and the simplest is a polynomial $p(t)$ that fits the given data as best as possible, that is $y_i = p(t_i)$. The process of finding $p(t)$ is called **polynomial interpolation**.

Note 1. Recall that we require $n + 1$ constants to describe a polynomial of degree n. Thus, given $n + 1$ data points, we look to construct a polynomial of degree n. As the following example shows, polynomial interpolation ultimately reduces to solving a system of equation.

Example 1. Suppose the ordered pair (t_i, y_i) records the sales of a company. The following sales record is known

$$(-1, 2), (0, 3), (1, 0), (3, 6)$$

The negative values of t_i may be interpreted as the years before the company is formally set up. So, we have the records of sales a year before the company is registered, at the end of the initial year and at the end of the second and fourth years. We would like to predict the sales at the end of 3rd year and at the end of the 9th month of the initial year.

Since 4 data points are given, we fit a polynomial of degree 3. Let $p(t) = c_0 + c_1 t + c_2 t^2 + c_3 t^3$ be the polynomial passing through the four given points. Since $p(-1) = 2$, $p(0) = 3$, $p(3) = 6$ and $p(3) = 6$, we have

$$\begin{aligned} c_0 - c_1 + c_2 - c_3 &= 2 \\ c_0 &= 3 \\ c_0 + c_1 + c_2 + c_3 &= 0 \\ c_0 + 3c_1 + 9c_2 + 27c_3 &= 6 \end{aligned}$$

We solve the system by Gauss elimination to obtain $c_0 = 3$, $c_1 = -2$, $c_2 = -2$ and $c_3 = 1$. The interpolating polynomial is $p(t) = 3 - 2t - 2t^2 + t^3$.

Now at the end of the third year, we predict the sales to be $p(2) = 3 - 4 - 8 + 8 = -1$ units. The 9th month in the initial year is $t = 0 + 9/12 = 3/4$ years of the company and the sales is $p(3/4)$ units.